Vacuum World AI Agent in Python

Description

This project simulates a simple vacuum cleaner agent operating in a two-room environment. It demonstrates core AI concepts such as state space representation, environment perception, and reflex agent behavior using Python. The agent perceives room status (clean or dirty) and decides when to clean or move, aiming to clean all rooms efficiently.

Features

- Two-room vacuum world environment simulation
- Reflex agent that acts based on current perceptions
- State encoding of vacuum location and room cleanliness status
- Console-based step-by-step visualization of agent actions

The program will simulate the vacuum agent cleaning both rooms, printing out the vacuum's location, room dirtiness, and chosen actions at each step until all rooms are clean.

Project Structure

- vacuum_world.py — main script containing environment and agent classes
- README.md — this documentation file

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